UPISC UNIVERSITY OF PITTSBURGH INFRASTRUCTURE SENSING COLLABORATION WORKSHOP

NATIONAL ENERGY TECHNOLOGY LABORATORY

Distributed Fiber Optics for Ultrasonic Guided Wave NDE for Spent Nuclear Fuel Canister and Buried Pipes

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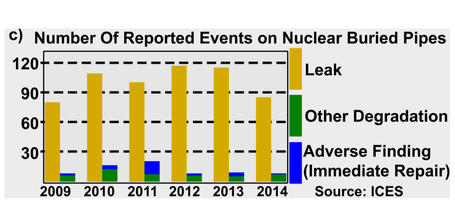
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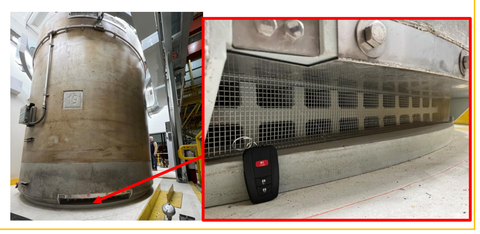
Motivation:

Current Status: *Buried pipelines and Dry Casks Storage Systems (DCSS) for nuclear waste storage poses challenges for traditional structural health monitoring methods due to hard and limited access, creating a demand for less intrusive techniques of easy installation. Current inspection methods practiced many times rely on visual inspections or highly costly operations to enable access to such structures. Meanwhile, the number of failure events on buried pipes and the growing figure of DCSS is alarming.*

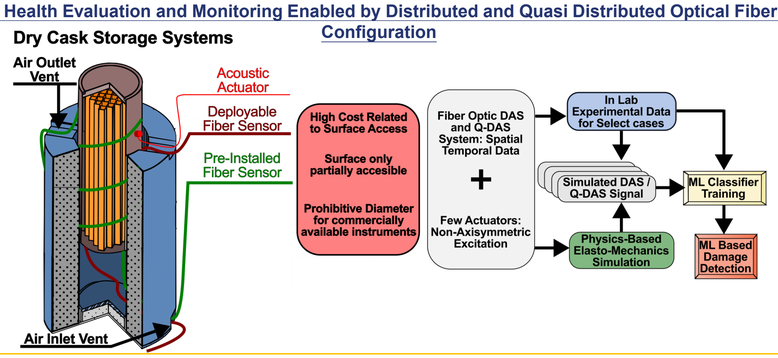
Objective: *Combine Ultrasonic Guided Waves (UGW) nondestructive evaluation (NDE) with distributed fiber optic sensing enhanced by artificial intelligence (AI) frameworks to quantitatively characterize the internal state of dry cask storage systems (DCSS). The methos pursued have the specific goal of:*

- *Facilitate inspection operation by adopting fiber sensors requiring single installation*
- *Enabling limited access inspections by reducing the number of required guided wave actuators*
- *Improve range of inspection of a single set of installed guided wave actuators*



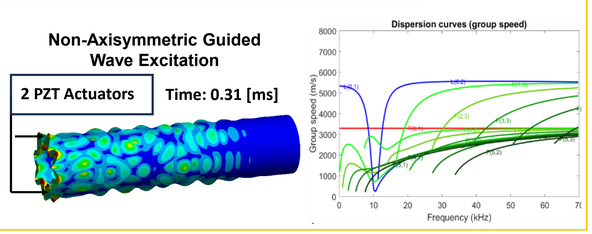


Structural Health Evaluation and Monitoring Enabled by Distributed and Quasi Distributed Optical Fiber Sensor Configuration



Ultrasonic Guided Wave (UGW) Non-Destructive Evaluation Method Principle

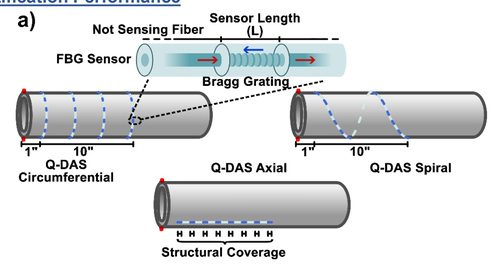
- An actuator (e.g. PZT) vibrates, generating a mechanical wave pulse
- The guided wave is reflected and refracted by mechanical discontinuities
- Guided waves can inspect longer distances due to lower attenuation
- Guided wave velocity is dependant on frequency



Investigation of Different Fiber Optic Sensor Installation Schemes on AI damage Identification Performance

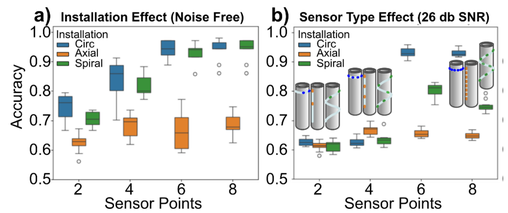
DFOS and Q-DFOS enable creative and optimized sampling of strain along the structure. How variations in parameters of sensor deployment can affect damage classification was studied by employing a simple Temporal Convolutional Neural Network classification model.

- Damage was modeled as notches at varying locations
- Simple straight forward fiber installation were considered
- Number of sampling points on the fiber was studied
- Gauge Length of distributed sensors is in the order of magnitude of UGW wavelength

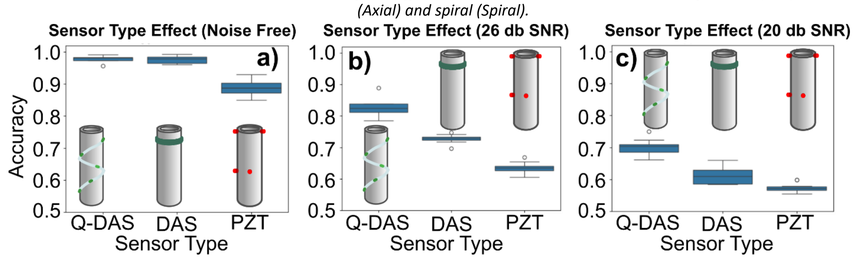


Schematic representation of straightforward possible fiber optic installations for Q-DAS and DAS systems

- Sampling location is a more impactful parameter than number of sampling points
- Axial points carry less unique information than points along circumference
- Combination of Axial and Circumferential points preserve unique features at large gauge lengths



Prediction accuracy deviation as a function of the number of sensors for 3 different sensors Q-DAS installations: Circumferential (Circ); Axial (Axial) and spiral (Spiral).

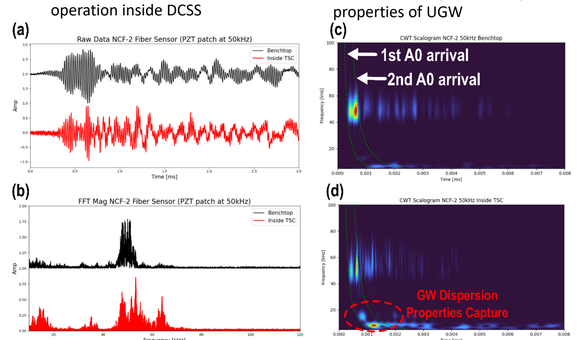


Prediction accuracy deviation for different types of sensor systems, considering Q-DAS, DAS and PZT arrangement with best performance

- When device limited, traditional PZT sampling approach at two axial locations is consistently outperformed
- Range of inspection and more detailed fiber installation must be taken into consideration and further studied

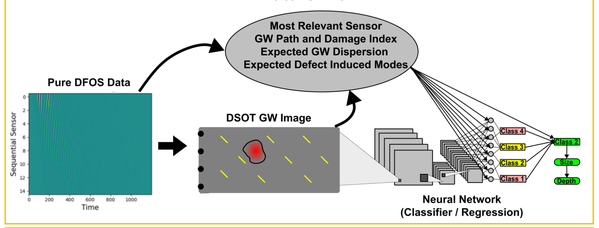
SMS Acoustics Sensor Testing for UGW NDE at Hanford Nuclear Facility

- Acoustic SMS sensors are quick to make in situ and highly sensitive
- Resisted to Installation and operation inside DCSS
- Frequency bandwidth sufficient to capture UGW
- Able to capture dispersion properties of UGW



Sensor data from inside HLC with UGW excited at 10kHz and 50kHz. (a) Timeseries data and (b) frequency spectrum (c) and (d) CWT spectrograms Green dashed lines represent expected A0 mode wavefront arrival time based on dispersion properties.

Future Work



References and Acknowledgements

[1] Sarcinelli et al. (2025) – *SPIE Optical Waveguide and Laser Sensors IV*
[2] Naeem et al. (2025) – *J. Lightwave Technol.*
[3] Zhang et al. (2025) – *Structural Health Monitoring*
[4] Sarcinelli et al. (2024) – *SPIE Optical Waveguide and Laser Sensors III*
[5] Naeem et al. (2025) – *SPIE Optical Waveguide and Laser Sensors IV*
[6] Naeem et al. (2024) – *SPIE Optical Waveguide and Laser Sensors III*
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